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Semiconductor Device and Method of Forming Encapsulated Vertical Interconnect Structure

Abstract

A semiconductor device has a first vertical interconnect substrate and a second vertical interconnect substrate. A first encapsulant is over the first vertical interconnect structure and second vertical interconnect structure. The first vertical interconnect structure is formed over a substrate. The first vertical interconnect structure can be a plurality of stacked vertical conductive posts. An electrical component is disposed between the first vertical interconnect structure and second vertical interconnect structure, or adjacent to the first vertical interconnect structure and second vertical interconnect structure. A second encapsulant is deposited around the electrical component. A portion of the second encapsulant is removed to expose a first surface of the first vertical interconnect structure. The substrate is removed to expose a second surface of the first vertical interconnect structure. A portion of the first encapsulant can be removed to expose a surface of the first vertical interconnect structure.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates in general to semiconductor devices and, more particularly, to a semiconductor device and method of forming an encapsulated vertical interconnect structure.

BACKGROUND OF THE INVENTION

[0002] Semiconductor devices are commonly found in modern electronic products. Semiconductor devices perform a wide range of functions, such as signal processing, high-speed calculations, transmitting and receiving electromagnetic signals, controlling electronic devices, photo-electric, and creating visual images for television displays. Semiconductor devices are found in the fields of communications, power conversion, networks, computers, entertainment, and consumer products. Semiconductor devices are also found in military applications, aviation, automotive, industrial controllers, and office equipment.

[0003] A semiconductor wafer typically contains a plurality of semiconductor die separated by a saw street. The semiconductor wafer is singulated into a plurality of individual semiconductor die. A semiconductor package may contain a semiconductor die mounted to an interconnect substrate or structure. The interconnect structure should provide vertical interconnect capability for the semiconductor die, sometimes to a significant height. The stacking of vias to form a high vertical interconnection can lead to irregular shapes and sizes of the vias, reduce manufacturing yield, high warpage, and increase defects. The vertical interconnect structure can have a limited height to diameter ratio, reducing interconnect density.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIGS. **1a-1c** illustrate a semiconductor wafer with a plurality of semiconductor die separated by a saw street;

[0005] FIGS. **2a-2g** illustrate a process of forming an encapsulated vertical interconnect structure;

[0006] FIGS. **3a-3d** illustrate an alternate process of forming a vertical interconnect structure for encapsulation;

[0007] FIGS. **4a-4h** illustrate packaging the encapsulated vertical interconnect structure from FIG. **2g** with an electrical component;

[0008] FIGS. **5a-5e** illustrate packaging the encapsulated vertical interconnect structure from FIG. **2f** with an electrical component;

[0009] FIGS. **6a-6e** illustrate a process of forming an encapsulated stacked vertical interconnect structure;

[0010] FIGS. **7a-7e** illustrate packaging the encapsulated stacked vertical interconnect structure from FIG. **6d** with an electrical component; and

[0011] FIG. **8** illustrates a printed circuit board (PCB) with different types of packages disposed on a surface of the PCB.

DETAILED DESCRIPTION OF THE DRAWINGS

[0012] The present invention is described in one or more embodiments in the following description

with reference to the figures, in which like numerals represent the same or similar elements. While the invention is described in terms of the best mode for achieving the invention's objectives, it will be appreciated by those skilled in the art that it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the invention as defined by the appended claims and their equivalents as supported by the following disclosure and drawings. The features shown in the figures are not necessarily drawn to scale. Elements having a similar function are assigned the same reference number in the figures. The term “semiconductor die” as used herein refers to both the singular and plural form of the words, and accordingly, can refer to both a single semiconductor device and multiple semiconductor devices.

[0013] Semiconductor devices are generally manufactured using two complex manufacturing processes: front-end manufacturing and back-end manufacturing. Front-end manufacturing involves the formation of a plurality of die on the surface of a semiconductor wafer. Each die on the wafer contains active and passive electrical components, which are electrically connected to form functional electrical circuits. Active electrical components, such as transistors and diodes, have the ability to control the flow of electrical current. Passive electrical components, such as capacitors, inductors, and resistors, create a relationship between voltage and current necessary to perform electrical circuit functions.

[0014] Back-end manufacturing refers to cutting or singulating the finished wafer into the individual semiconductor die and packaging the semiconductor die for structural support, electrical interconnect, and environmental isolation. To singulate the semiconductor die, the wafer is scored and broken along non-functional regions of the wafer called saw streets or scribes. The wafer is singulated using a laser cutting tool or saw blade. After singulation, the individual semiconductor die are disposed on a package substrate that includes pins or contact pads for interconnection with other system components. Contact pads formed over the semiconductor die are then connected to contact pads within the package. The electrical connections can be made with conductive layers, bumps, stud bumps, conductive paste, or wirebonds. An encapsulant or other molding material is deposited over the package to provide physical support and electrical isolation. The finished package is then inserted into an electrical system and the functionality of the semiconductor device is made available to the other system components.

[0015] FIG. **1a** shows a semiconductor wafer **100** with a base substrate material **102**, such as silicon, germanium, aluminum phosphide, aluminum arsenide, gallium arsenide, gallium nitride, indium phosphide, silicon carbide, or other bulk material for structural support. A plurality of semiconductor die or electrical components **104** is formed on wafer **100** separated by a non-active, inter-die wafer area or saw street **106**. Saw street **106** provides cutting areas to singulate semiconductor wafer **100** into individual semiconductor die **104**. In one embodiment, semiconductor wafer **100** has a width or diameter of 100-450 millimeters (mm).

[0016] FIG. **1b** shows a cross-sectional view of a portion of semiconductor wafer **100**. Each semiconductor die **104** has a back or non-active surface **108** and an active surface **110** containing analog or digital circuits implemented as active devices, passive devices, conductive layers, and dielectric layers formed within the die and electrically interconnected according to the electrical design and function of the die. For example, the circuit may include one or more transistors, diodes, and other circuit elements formed within active surface **110** to implement analog circuits or digital circuits, such as digital signal processor (DSP), application specific integrated circuits (ASIC), memory, or other signal processing circuit. Semiconductor die **104** may also contain IPDs, such as inductors, capacitors, and resistors, for RF signal processing.

[0017] An electrically conductive layer **112** is formed over active surface **110** using physical vapor deposition (PVD), chemical vapor deposition (CVD), electrolytic plating, electroless plating process, or other suitable metal deposition process. Conductive layer **112** can be one or more layers of aluminum (Al), copper (Cu), tin (Sn), nickel (Ni), gold (Au), silver (Ag), or other suitable electrically conductive material. Conductive layer **112** operates as contact pads electrically

connected to the circuits on active surface **110**.

[0018] In FIG. **1c**, semiconductor wafer **100** is singulated through saw street **106** using a saw blade or laser cutting tool **119** into individual semiconductor die **104**. The individual semiconductor die **104** can be inspected and electrically tested for identification of known good die or unit (KGD/KGU) post singulation.

[0019] FIGS. **2a-2g** illustrate a process of forming an encapsulated vertical interconnect structure. FIG. **2a** shows a temporary substrate **120** made with base material **122** as a multi-layer flexible laminate, ceramic, copper clad laminate (CCL), glass, or epoxy molding compound. Base material **122** can contain one or more layers of SiO₂, Si₃N₄, SiON, Ta₂O₅, Al₂O₃, solder resist, polyimide, BCB, PBO, and other material having similar insulating and structural properties. Substrate **120** may include one or more laminated layers of polytetrafluoroethylene (PTFE) pre-impregnated (prepreg), FR-4, FR-1, CEM-1, or CEM-3 with a combination of phenolic cotton paper, epoxy, resin, woven glass, matte glass, polyester, and other reinforcement fibers or fabrics. Substrate **120** can be silicon, germanium, aluminum phosphide, aluminum arsenide, gallium arsenide, gallium nitride, indium phosphide, silicon carbide, or other bulk material for structural support. In another embodiment, substrate **120** can also be any suitable laminate interposer, PCB, wafer-form, strip interposer, leadframe, or other type of substrate. Substrate **120** has a thickness T of 100 μm or more micrometers (μm) and includes first major surface **124** and second major surface **126** opposite surface **124**.

[0020] In FIG. **2b**, electrically conductive posts or pillars or columns **130a**, **130b**, **130c**, **130d**, and **130e** are formed on surface **124**. Conductive posts **130a-130e** can be one or more layers of Al, Cu, Sn, Ni, Au, Ag, or other suitable electrically conductive material. In one embodiment, conductive posts **130a-130e** are attached to substrate **120** with conductive paste or solder **132**. Bonding material **132** can also be a conductive or non-conductive adhesive thin-film or die attach film (DAF) to form a vertically standing structure of conductive posts **130a** to **130e** on surface **124** of substrate **120**. FIG. **2c** shows conductive posts **130a-130e** attached to substrate **120** as vertical interconnect structure **136**. In one embodiment, conductive posts **130a-130e** have a height H1 of about 100 μm and diameter D1 of 50 μm, depending on the desired pitch manufacturing process.

[0021] In another embodiment, and continuing from FIG. **2a**, photoresist layer **140** is formed over surface **124** of substrate **120**, as shown in FIG. **3a**. In FIG. **3b**, photoresist layer **140** is patterned and etched to form openings **142**. Alternatively, openings **142** can be formed by laser direct ablation (LDA) using laser **144**.

[0022] In FIG. **3c**, electrically conductive material **146** is deposited into openings **142** by PVD, CVD, electrolytic plating, electroless plating process, or other suitable metal deposition process. Conductive material **146** can be one or more layers of Al, Cu, Sn, Ni, Au, Ag, or other suitable electrically conductive material.

[0023] In FIG. **3d**, photoresist layer **140** is removed leaving conductive posts or columns or pillars **148a**, **148b**, **148c**, **148d**, and **148e** formed over surface **124** of substrate **120**, as in vertical interconnect structure **150**, similar to vertical interconnect structure **136**.

[0024] Returning to FIG. **2d**, an encapsulant or molding compound **152** is deposited over and around conductive posts **130a-130e** and surface **124** of substrate **120** in vertical interconnect structure **136** using a paste printing, compressive molding, transfer molding, liquid encapsulant molding, vacuum lamination, spin coating, or other suitable applicator. Encapsulant **152** can be polymer composite material, such as epoxy resin with filler, epoxy acrylate with filler, or polymer with proper filler. Encapsulant **152** is non-conductive, provides structural support, and environmentally protects the semiconductor device from external elements and contaminants. Although the subsequent process steps are described in terms of vertical interconnect structure **136**, a similar encapsulation would apply to vertical interconnect structure **150** if that embodiment is utilized.

[0025] In FIG. **2e**, a portion of encapsulant **152** is removed by grinder **156** to planarize surface **158**

of encapsulant **152** and expose surface **154** of conductive pillars **130a-130e**. FIG. **2f** shows encapsulated vertical interconnect structure **160** post grinding with planarized surface **158** and exposed surface **154** of conductive posts **130a-130e**.

[0026] In another embodiment, continuing from FIG. **2d**, substrate **120** is removed by chemical mechanical polishing (CMP), mechanical peel-off, mechanical grinding, thermal bake, UV light, or wet stripping, leaving encapsulated vertical interconnect structure **166** with encapsulant **152** covering conductive posts **130a-130e**, while planarizing surface **169** of encapsulant **152** and exposing surface **167** of the conductive posts, as shown in FIG. **2g**.

[0027] FIG. **4a** shows a temporary substrate **170** made with base material **171** as a multi-layer flexible laminate, ceramic, CCL, glass, or epoxy molding compound. Base material **171** can contain one or more layers of SiO₂, Si₃N₄, SiON, Ta₂O₅, Al₂O₃, solder resist, polyimide, BCB, PBO, and other material having similar insulating and structural properties. Substrate **170** may include one or more laminated layers of PTFE prepreg, FR-4, FR-1, CEM-1, or CEM-3 with a combination of phenolic cotton paper, epoxy, resin, woven glass, matte glass, polyester, and other reinforcement fibers or fabrics. In another embodiment, substrate **120** can also be any suitable laminate interposer, PCB, wafer-form, strip interposer, leadframe, or other type of substrate. Substrate **170** has a thickness T and includes first major surface **172** and second major surface **174** opposite surface **172**.

[0028] Encapsulated vertical interconnect structures **166a** and **166b** from FIG. **2g** and electrical component **168** are disposed over surface **172** of substrate **170**. For example, electrical component **168** can be similar to, or made similar to, semiconductor die **104** from FIG. **1c** with conductive layer **112** oriented toward surface **172**. Electrical component **168** can be other semiconductor die, semiconductor packages, surface mount devices, RF components, discrete electrical devices, or integrated passive devices (IPD).

[0029] Encapsulated vertical interconnect structures **166a-166b** and electrical component **168** are positioned over substrate **170** using a pick and place operation. Electrical component **168** is disposed between encapsulated vertical interconnect structures **166a-166b**, or adjacent to either encapsulated vertical interconnect structure **166a** or **166b**. Encapsulated vertical interconnect structures **166a-166b** and electrical component **168** can be bonded to surface **172** with a bonding layer (not shown). FIG. **4b** shows encapsulated vertical interconnect structures **166a-166b** and electrical component **168** bonded to substrate **170**.

[0030] In FIG. **4c**, an encapsulant or molding compound **178** is deposited over and around encapsulated vertical interconnect structures **166a-166b** and electrical component **168** and surface **172** of substrate **170** using a paste printing, compressive molding, transfer molding, liquid encapsulant molding, vacuum lamination, spin coating, or other suitable applicator. Encapsulant **178** can be polymer composite material, such as epoxy resin with filler, epoxy acrylate with filler, or polymer with proper filler. Encapsulant **178** is non-conductive, provides structural support, and environmentally protects the semiconductor device from external elements and contaminants.

[0031] In FIG. **4d**, a portion of encapsulant **178** is removed by grinder **180** to planarize surface **182** of encapsulant **178** and expose surface **183** of conductive pillars **130a-130e**. FIG. **2e** shows the encapsulated vertical interconnect structure post grinding with planarized surface **182** and exposed surface **194** of conductive posts **130a-130e**. Surface **182** is planar with a surface of encapsulant **178**. Encapsulant **178** may or may not cover back surface **108** of electrical component **168**. FIG. **4f** is a top view of encapsulated vertical interconnect structures **166a-166b** and electrical component **168** bonded to substrate **170**.

[0032] In FIG. **4g**, substrate **170** is removed by CMP, mechanical peel-off, mechanical grinding, thermal bake, UV light, or wet stripping, leaving semiconductor package **185**.

[0033] As an optional feature, a build-up interconnect structure **184** is formed over semiconductor package **185**, as in FIG. **4h**. Conductive layers **186** and insulating layers **188** are formed within interconnect substrate **184**. Conductive layers **186** can be one or more layers of Al, Cu, Sn, Ni, Au,

Ag, or other suitable electrically conductive material. Conductive layers **186** can be formed using PVD, CVD, electrolytic plating, electroless plating process, or other suitable metal deposition process. Conductive layer **186** provides horizontal electrical interconnect across substrate **184** and vertical electrical interconnect through the substrate. Portions of conductive layers **186** can be electrically common or electrically isolated depending on the design and function of electrical component **168**. Insulating layers **188** contain one or more layers of SiO₂, Si₃N₄, SiON, Ta₂O₅, Al₂O₃, solder resist, polyimide, BCB, PBO, and other material having similar insulating and structural properties. Insulating layers **188** can be formed using PVD, CVD, printing, lamination, spin coating, spray coating, sintering or thermal oxidation. Insulating layers **188** provide isolation between conductive layers **186**. There can be multiple conductive layers like **186** separated by multiple insulating layers like **188**.

[0034] An electrically conductive bump material is deposited over conductive layer **186** using an evaporation, electrolytic plating, electroless plating, ball drop, or screen printing process. The bump material can be Al, Sn, Ni, Au, Ag, Pb, Bi, Cu, solder, and combinations thereof, with an optional flux solution. For example, the bump material can be eutectic Sn/Pb, high-lead solder, or lead-free solder. The bump material is bonded to conductive layer **186** using a suitable attachment or bonding process. In one embodiment, the bump material is reflowed by heating the material above its melting point to form balls or bumps **189**. In one embodiment, bump **189** is formed over an under bump metallization (UBM) having a wetting layer, barrier layer, and adhesive layer. Bump **189** can also be compression bonded or thermocompression bonded to conductive layer **186**. Bump **189** represents one type of interconnect structure that can be formed over conductive layer **186**. The interconnect structure can also use bond wires, conductive paste, stud bump, micro bump, or other electrical interconnect.

[0035] Semiconductor package **185** or **190** represents an encapsulated vertical interconnect structure applicable to wafer level processing (WLP). Conductive posts **130** or **148** provide high post height to diameter ratio. Encapsulated vertical interconnect structures **160** and **166** are pre-formed over substrate **120** prior to mounting to substrate **170**. Encapsulated vertical structures **160** and **166** are lamination free so there is little or no risk of delamination. Conductive posts **130** can also be glass or plastic fiber to transmit light in optical applications.

[0036] In another embodiment, FIG. 5a shows encapsulated vertical interconnect structures **160a** and **160b** from FIG. 2f and electrical component **168** disposed over surface **172** of substrate **170**. Components having a similar function are assigned the same reference number. Encapsulated vertical interconnect structures **160a-160b** and electrical component **168** are positioned over substrate **170** using a pick and place operation, similar to FIG. 4a-4b. In this case, temporary substrate **120** of encapsulated vertical interconnect structures **160a-160b** is oriented away from surface **172**. Electrical component **168** is disposed between encapsulated vertical interconnect structures **160a-160b**, or adjacent to either encapsulated vertical interconnect structure **160a** or **160b**. Encapsulated vertical interconnect structures **160a-160b** and electrical component **168** can be bonded to surface **172** with a bonding layer.

[0037] In FIG. 5b, an encapsulant or molding compound **192** is deposited over and around encapsulated vertical interconnect structures **160a-160b** and electrical component **168** and surface **172** of substrate **170** using a paste printing, compressive molding, transfer molding, liquid encapsulant molding, vacuum lamination, spin coating, or other suitable applicator. Encapsulant **192** can be polymer composite material, such as epoxy resin with filler, epoxy acrylate with filler, or polymer with proper filler. Encapsulant **192** is non-conductive, provides structural support, and environmentally protects the semiconductor device from external elements and contaminants.

[0038] In FIG. 5c, a portion of encapsulant **178** and the entirety of substrate **120** are removed by a grinder, similar to FIG. 4e, to planarize surface **191** of encapsulant **192** and expose surface **193** of conductive pillars **130a-130e**. Surface **191** is planar with a surface of encapsulant **192**. A top view of encapsulated vertical interconnect structures **160a-160b** and electrical component **168** bonded to

substrate **170** would be similar to FIG. **4f**.

[0039] In FIG. **5d**, substrate **170** is removed by CMP, mechanical peel-off, mechanical grinding, thermal bake, UV light, or wet stripping, leaving semiconductor package **195**.

[0040] As an optional feature, a build-up interconnect structure **194** is formed over semiconductor package **195**, as in FIG. **5e**. Conductive layers **196** and insulating layers **198** are formed within interconnect substrate **194**. Conductive layers **196** can be one or more layers of Al, Cu, Sn, Ni, Au, Ag, or other suitable electrically conductive material. Conductive layers **196** can be formed using PVD, CVD, electrolytic plating, electroless plating process, or other suitable metal deposition process. Conductive layer **196** provides horizontal electrical interconnect across substrate **194** and vertical electrical interconnect through the substrate. Portions of conductive layers **196** can be electrically common or electrically isolated depending on the design and function of electrical component **168**. Insulating layers **198** contain one or more layers of SiO₂, Si₃N₄, SiON, Ta₂O₅, Al₂O₃, solder resist, polyimide, BCB, PBO, and other material having similar insulating and structural properties. Insulating layers **198** can be formed using PVD, CVD, printing, lamination, spin coating, spray coating, sintering or thermal oxidation. Insulating layers **198** provide isolation between conductive layers **186**. There can be multiple conductive layers like **196** separated by multiple insulating layers like **198**.

[0041] An electrically conductive bump material is deposited over conductive layer **196** using an evaporation, electrolytic plating, electroless plating, ball drop, or screen printing process. The bump material can be Al, Sn, Ni, Au, Ag, Pb, Bi, Cu, solder, and combinations thereof, with an optional flux solution. For example, the bump material can be eutectic Sn/Pb, high-lead solder, or lead-free solder. The bump material is bonded to conductive layer **196** using a suitable attachment or bonding process. In one embodiment, the bump material is reflowed by heating the material above its melting point to form balls or bumps **199**. In one embodiment, bump **199** is formed over a UBM having a wetting layer, barrier layer, and adhesive layer. Bump **199** can also be compression bonded or thermocompression bonded to conductive layer **196**. Bump **199** represents one type of interconnect structure that can be formed over conductive layer **196**. The interconnect structure can also use bond wires, conductive paste, stud bump, micro bump, or other electrical interconnect.

[0042] Semiconductor package **200** represents an encapsulated vertical interconnect structure applicable to WLP. Conductive posts **130** or **148** provide high post height to diameter ratio. Encapsulated vertical interconnect structures **160** and **166** are pre-formed over substrate **120** prior to mounting to substrate **170**. Encapsulated vertical structures **160** and **166** are lamination free so there is little or no risk of delamination. Conductive posts **130** can also be glass or plastic fiber to transmit light in optical applications.

[0043] In another embodiment, the conductive posts like **130a-130e** can be stacked. FIG. **6a** shows vertical interconnect structure **136** from FIG. **2c**, although vertical interconnect structure **150** from FIG. **3d** can be used. Vertical interconnect structure **136a** is disposed over vertical interconnect structure **136b** using a pick and place operation. Conductive posts **130a-130e** in vertical interconnect structure **136a** are brought into contact with conductive posts **130a-130e** in vertical interconnect structure **136b** and bonded with conductive paste or solder **202**, as vertical interconnect structure **210** in FIG. **6b**. In one embodiment, the stacked conductive posts **211** have a height H₂ of 2*H₁ and diameter D₂ of about the same or larger than D₁ to increase the height to diameter aspect ratio suitable for embedded semiconductor die **104** or Si photonic component required a thickness more or equal to 100 μm.

[0044] In FIG. **6c**, an encapsulant or molding compound **212** is deposited over and around vertical interconnect structure **210** using a paste printing, compressive molding, transfer molding, liquid encapsulant molding, vacuum lamination, spin coating, or other suitable applicator. Encapsulant **212** can be polymer composite material, such as epoxy resin with filler, epoxy acrylate with filler, or polymer with proper filler. Encapsulant **212** is non-conductive, provides structural support, and environmentally protects the semiconductor device from external elements and contaminants.

[0045] In FIG. 6d, a portion of encapsulant **212** and the entirety of substrate **120** in vertical interconnect structure **136a** are removed by grinder **214**, similar to FIG. 4e, to planarize surface **218** of encapsulant **192** and expose surface **216** of conductive pillars **130a-130e** in vertical interconnect structure **136a**. FIG. 6e shows encapsulated vertical interconnect structure **220** post grinding with planarized surface **218** and exposed conductive posts **130a-130e**.

[0046] FIG. 7a shows a temporary substrate **230** made with base material **231** as a multi-layer flexible laminate, ceramic, CCL, glass, or epoxy molding compound. Base material **231** can contain one or more layers of SiO₂, Si₃N₄, SiON, Ta₂O₅, Al₂O₃, solder resist, polyimide, BCB, PBO, and other material having similar insulating and structural properties. Substrate **230** may include one or more laminated layers of PTFE prepreg, FR-4, FR-1, CEM-1, or CEM-3 with a combination of phenolic cotton paper, epoxy, resin, woven glass, matte glass, polyester, and other reinforcement fibers or fabrics. In another embodiment, substrate **230** can also be any suitable laminate interposer, PCB, wafer-form, strip interposer, leadframe, or other type of substrate. Substrate **230** has a thickness T and includes first major surface **232** and second major surface **234** opposite surface **232**. Encapsulated vertical interconnect structures **220a-220b** from FIG. 6e and electrical component **168** are positioned over substrate **230** using a pick and place operation, similar to FIG. 4a-4b. In this case, the remaining temporary substrate **120** of encapsulated vertical interconnect structures **220a-220b** is oriented away from surface **232**. Electrical component **168** is disposed between encapsulated vertical interconnect structures **220a-220b**, or adjacent to either encapsulated vertical interconnect structure **220a** or **220b**. Encapsulated interconnect structures **220a-220b** and electrical component **168** can be bonded to surface **232** with a bonding layer.

[0047] In FIG. 7b, an encapsulant or molding compound **236** is deposited over and around encapsulated vertical interconnect structures **220a-220b** and electrical component **168** and surface **232** of substrate **230** using a paste printing, compressive molding, transfer molding, liquid encapsulant molding, vacuum lamination, spin coating, or other suitable applicator. Encapsulant **236** can be polymer composite material, such as epoxy resin with filler, epoxy acrylate with filler, or polymer with proper filler. Encapsulant **236** is non-conductive, provides structural support, and environmentally protects the semiconductor device from external elements and contaminants.

[0048] In FIG. 7c, a portion of encapsulant **236** and the entirety of substrate **120** of vertical interconnect structure **136a** are removed by a grinder, similar to FIG. 4e, to planarize surface **238** of encapsulant **236** and expose surface **239** of conductive pillars **130a-130e**. Surface **238** is planar with a surface of encapsulant **236**. A top view of encapsulated vertical interconnect structures **220a-220b** and electrical component **168** bonded to substrate **230** would be similar to FIG. 4f.

[0049] In FIG. 7d, substrate **230** is removed by CMP, mechanical peel-off, mechanical grinding, thermal bake, UV light, or wet stripping, leaving semiconductor package **241**.

[0050] As an optional feature, a build-up interconnect structure **240** is formed over semiconductor package **241**, as in FIG. 7e. Conductive layers **242** and insulating layers **244** are formed within interconnect substrate **240**. Conductive layers **242** can be one or more layers of Al, Cu, Sn, Ni, Au, Ag, or other suitable electrically conductive material. Conductive layers **242** can be formed using PVD, CVD, electrolytic plating, electroless plating process, or other suitable metal deposition process. Conductive layer **242** provides horizontal electrical interconnect across substrate **240** and vertical electrical interconnect through the substrate. Portions of conductive layers **242** can be electrically common or electrically isolated depending on the design and function of electrical component **168**. Insulating layers **244** contain one or more layers of SiO₂, Si₃N₄, SiON, Ta₂O₅, Al₂O₃, solder resist, polyimide, BCB, PBO, and other material having similar insulating and structural properties. Insulating layers **244** can be formed using PVD, CVD, printing, lamination, spin coating, spray coating, sintering or thermal oxidation. Insulating layers **244** provide isolation between conductive layers **242**. There can be multiple conductive layers like **242** separated by multiple insulating layers like **244**.

[0051] An electrically conductive bump material is deposited over conductive layer **242** using an

evaporation, electrolytic plating, electroless plating, ball drop, or screen printing process. The bump material can be Al, Sn, Ni, Au, Ag, Pb, Bi, Cu, solder, and combinations thereof, with an optional flux solution. For example, the bump material can be eutectic Sn/Pb, high-lead solder, or lead-free solder. The bump material is bonded to conductive layer **242** using a suitable attachment or bonding process. In one embodiment, the bump material is reflowed by heating the material above its melting point to form balls or bumps **248**. In one embodiment, bump **248** is formed over a UBM having a wetting layer, barrier layer, and adhesive layer. Bump **248** can also be compression bonded or thermocompression bonded to conductive layer **242**. Bump **248** represents one type of interconnect structure that can be formed over conductive layer **242**. The interconnect structure can also use bond wires, conductive paste, stud bump, micro bump, or other electrical interconnect. [0052] Semiconductor package **241** or **250** represents an encapsulated vertical interconnect structure applicable to WLP. Conductive posts **211** provide a higher post height to diameter ratio by nature of being stacked. Encapsulated vertical interconnect structures **160** and **166** are pre-formed over substrate **120** prior to mounting to substrate **170**. Encapsulated vertical structures **160** and **166** are lamination free so there is little or no risk of delamination. Conductive posts **211** can also be glass or plastic fiber to transmit light in optical applications.

[0053] FIG. **8** illustrates electrical device **400** having a chip carrier substrate or PCB **402** with a plurality of semiconductor packages disposed on a surface of PCB **402**, including semiconductor packages **185**, **190**, **195**, **200**, **241**, and **250**. Electrical device **400** can have one type of semiconductor package, or multiple types of semiconductor packages, depending on the application.

[0054] Electrical device **400** can be a stand-alone system that uses the semiconductor packages to perform one or more electrical functions. Alternatively, electrical device **400** can be a subcomponent of a larger system. For example, electrical device **400** can be part of a tablet, cellular phone, digital camera, communication system, or other electrical device. Alternatively, electrical device **400** can be a graphics card, network interface card, or other signal processing card that can be inserted into a computer. The semiconductor package can include microprocessors, memories, ASIC, logic circuits, analog circuits, RF circuits, discrete devices, or other semiconductor die or electrical components. Miniaturization and weight reduction are essential for the products to be accepted by the market. The distance between semiconductor devices may be decreased to achieve higher density.

[0055] In FIG. **8**, PCB **402** provides a general substrate for structural support and electrical interconnect of the semiconductor packages disposed on the PCB. Conductive signal traces **404** are formed over a surface or within layers of PCB **402** using evaporation, electrolytic plating, electroless plating, screen printing, or other suitable metal deposition process. Signal traces **404** provide for electrical communication between each of the semiconductor packages, mounted components, and other external system components. Traces **404** also provide power and ground connections to each of the semiconductor packages.

[0056] In some embodiments, a semiconductor device has two packaging levels. First level packaging is a technique for mechanically and electrically attaching the semiconductor die to an intermediate substrate. Second level packaging involves mechanically and electrically attaching the intermediate substrate to the PCB. In other embodiments, a semiconductor device may only have the first level packaging where the die is mechanically and electrically disposed directly on the PCB. For the purpose of illustration, several types of first level packaging, including bond wire package **406** and flipchip **408**, are shown on PCB **402**. Additionally, several types of second level packaging, including ball grid array (BGA) **410**, bump chip carrier (BCC) **412**, land grid array (LGA) **416**, multi-chip module (MCM) or SIP module **418**, quad flat non-leaded package (QFN) **420**, quad flat package **422**, embedded wafer level ball grid array (eWLB) **424**, and wafer level chip scale package (WLCSP) **426** are shown disposed on PCB **402**. In one embodiment, eWLB **424** is a fan-out wafer level package (Fo-WLP) and WLCSP **426** is a fan-in wafer level package (Fi-

WLP). Depending upon the system requirements, any combination of semiconductor packages, configured with any combination of first and second level packaging styles, as well as other electrical components, can be connected to PCB 402. In some embodiments, electrical device 400 includes a single attached semiconductor package, while other embodiments call for multiple interconnected packages. By combining one or more semiconductor packages over a single substrate, manufacturers can incorporate pre-made components into electrical devices and systems. Because the semiconductor packages include sophisticated functionality, electrical devices can be manufactured using less expensive components and a streamlined manufacturing process. The resulting devices are less likely to fail and less expensive to manufacture resulting in a lower cost for consumers.

[0057] While one or more embodiments of the present invention have been illustrated in detail, the skilled artisan will appreciate that modifications and adaptations to those embodiments may be made without departing from the scope of the present invention as set forth in the following claims.

Claims

1. A method of making a semiconductor device, comprising: forming a first vertical interconnect substrate; forming a second vertical interconnect substrate; depositing a first encapsulant over the first vertical interconnect structure and second vertical interconnect structure; and disposing an electrical component between the first vertical interconnect structure and second vertical interconnect structure.
2. The method of claim 1, further including depositing a second encapsulant around the electrical component.
3. The method of claim 2, further including removing a portion of the second encapsulant to expose a surface of the first vertical interconnect structure.
4. The method of claim 1, further including forming the first vertical interconnect structure over a substrate.
5. The method of claim 4, further including removing the substrate to expose a surface of the first vertical interconnect structure.
6. The method of claim 1, further including removing a portion of the first encapsulant to expose the first vertical interconnect structure.
7. A method of making a semiconductor device, comprising: forming a vertical interconnect substrate; depositing a first encapsulant over the vertical interconnect structure; and disposing an electrical component adjacent to the vertical interconnect structure.
8. The method of claim 7, further including depositing a second encapsulant around the electrical component.
9. The method of claim 8, further including removing a portion of the second encapsulant to expose a surface of the vertical interconnect structure.
10. The method of claim 7, further including forming the vertical interconnect structure over a substrate.
11. The method of claim 10, further including removing the substrate to expose a surface of the vertical interconnect structure.
12. The method of claim 7, further including removing a portion of the first encapsulant to expose the vertical interconnect structure.
13. The method of claim 7, further including forming a build-up interconnect structure over the vertical interconnect structure.
14. A semiconductor device, comprising: a first vertical interconnect substrate; a second vertical interconnect substrate; a first encapsulant deposited over the first vertical interconnect structure and second vertical interconnect structure; and an electrical component disposed between the first vertical interconnect structure and second vertical interconnect structure.

- 15.** The semiconductor device of claim 14, further including a second encapsulant deposited around the electrical component.
 - 16.** The semiconductor device of claim 15, wherein a surface of the first vertical interconnect structure is planar with a surface of the second encapsulant.
 - 17.** The semiconductor device of claim 14, wherein the first vertical interconnect structure includes a plurality of stacked vertical conductive posts.
 - 18.** The semiconductor device of claim 14, further including a build-up interconnect structure formed over the first vertical interconnect structure.
 - 19.** The semiconductor device of claim 14, further including a substrate, wherein the first vertical interconnect structure is formed over the substrate.
 - 20.** A semiconductor device, comprising: a vertical interconnect substrate; a first encapsulant deposited over the vertical interconnect structure; and an electrical component disposed adjacent to the vertical interconnect structure.
 - 21.** The semiconductor device of claim 20, further including a second encapsulant deposited around the electrical component.
 - 22.** The semiconductor device of claim 21, wherein a surface of the first vertical interconnect structure is planar with a surface of the second encapsulant.
 - 23.** The semiconductor device of claim 20, wherein the first vertical interconnect structure includes a plurality of stacked vertical conductive posts.
 - 24.** The semiconductor device of claim 20, further including a build-up interconnect structure formed over the first vertical interconnect structure.
 - 25.** The semiconductor device of claim 20, further including a substrate, wherein the first vertical interconnect structure is formed over the substrate.
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